

Certified Temporal Frontiers for Aggregate Queries over Hidden Matchings and Privacy-Coarsened Attributes

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ABSTRACT

Public releases often preserve one row per transaction while suppressing the key that joins rows into events and coarsening sensitive endpoint attributes. Downstream aggregate queries then depend jointly on a latent relation and the attributes attached to its endpoints. We optimize a query over feasible worlds that jointly assign a nonbipartite core–buffer matching and categorical labels subject to released supports and threshold-count factors. Independent label supports collapse to weighted matching; coupled worlds need not. We develop an exact temporal-frontier dynamic program whose state retains only live labels, matching bits, active capped-count automata, an omitted-edge budget, and a capped integer score resource. It returns attained lower and upper endpoints with witness worlds. With w live records, compiled support d , at most r active threshold factors of cap k , omission budget Γ , and integer score target b , the live-key bound is $(2d)^w [3(k+1)]^r (\Gamma+1)(b+1)$. The dependence on b is substantive: the associated threshold decision problem is weakly NP-complete even on disjoint four-cycles of path-width two. A downward score quantization gives certified outward query endpoints while relaxing the original score by at most $\eta|C|$; it is a bicriteria certificate, not a query FPTAS. We use matching-level split conformal prediction as a known safety layer, with one score normalization frozen across the entire candidate-omission path. A locked benchmark matches every applicable exhaustive or analytic reference, exposes order-dependent width, and shows the pseudo-polynomial score frontier; it makes no production-speed claim. Chicago ride pooling is a gated flagship application: realized co-presence is released, while partner identity and fine geography are hidden. Current City documentation supports a threshold-and-paired-end coarsening rule, but complete-day estimates remain conditional on run closure, implementation validation, and release-snapshot stability.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning**; • **Information systems** → *Data provenance*; • **Theory of computation** → *Design and analysis of algorithms*.

KEYWORDS

hidden matching, exact dynamic programming, aggregate bounds, privacy coarsening

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1 INTRODUCTION

Many public datasets preserve outcomes for individual records but remove the key identifying which records participated in the same event. Sensitive endpoint attributes may also be generalized or suppressed. Aggregate quantities such as the composition of matched pairs, referral chains, or shared-resource groups then depend on two latent objects at once: the relation and the attributes attached to its endpoints. Selecting one linkage or deleting rows with missing context creates precision by replacing released uncertainty with analyst certainty.

Propagating linkage uncertainty alone does not resolve this problem. Candidate edges can interact with latent endpoint attributes, privacy bucket counts make attribute completions mutually dependent, and the hidden matching determines which endpoint attributes enter the query. Separate link and attribute bounds can therefore combine marginally admissible choices that no single released database could have produced.

We represent the release by a set of feasible worlds. Each world jointly specifies a hidden matching and latent endpoint attributes, satisfies every released value and global coarsening constraint, and covers every core record exactly once. We compute attained lower and upper endpoints of a declared aggregate query over this set. These endpoints are conditional on the candidate supergraph and observation operator; they are not a point reconstruction, and the closed interval between them need not consist entirely of attainable values.

Learning is optional and coverage-audited. An edge model supplies a compatibility score, but node outcomes alone do not identify partner probabilities. We restrict feasible worlds by matching-level normalized regret and calibrate the radius on independent markets with observed relations. A separate supergraph budget permits selected edges outside a baseline graph. Thus learned sharpening and candidate-support sensitivity remain visible rather than being folded into one asserted graph.

Chicago ride pooling motivates the full structure. Consider a boundary-complete, internally consistent cohort with a positive shared-match flag and pooled chain size two. The public record reports realized co-presence but not the partner key. Fine geography can also be suppressed. The eventual application bounds neighborhood-context composition among those customer transactions. It does not infer rider socioeconomic status, preferences, durable ties, or all shared runs.

This working version reports three completed results and artifacts. First, it defines joint feasible-world query semantics and identifies the exact collapse to weighted matching. Second, it develops and implements a score-aware temporal frontier algorithm for core–buffer matching, compiled labels, active release counts, and a global budget for omitted candidate edges; it proves endpoint and witness recovery, the pseudo-polynomial score boundary, and a certified outward score relaxation. A provenance-bearing compiler

maps explicit DNF release implications to replayable states without changing their projected world set. Third, we supply independently implemented exhaustive checks, a bounded capacity profile, and external real/synthetic relation-truth stress tests. The real relation is deliberately reported as nonmatching rather than forced into the model. Explicit-DNF label duplication and the vertex-separation identity are audit and implementation devices, not knowledge-compilation novelties. Matching-level conformal calibration is an adopted safety layer, not a novelty claim. The full external corpus, a boundary-complete Chicago run, and production-scale schedules remain explicit KDD submission gates.

2 RELATED WORK

Uncertain linkage and aggregate queries. Globally compatible record linkages and downstream uncertainty are established objects. Sadinle models bipartite linkage as a constrained matching [21]; Hua and Pei evaluate aggregate queries over compatible probabilistic linkages [14]; and Turkcapar and Krishnan compute extremal aggregate answers over candidate-constrained two-table linkages [27]. Attribute-annotated uncertain databases propagate value bounds through relational queries [11]. Our target difference is conditional and narrower: a within-file nonbipartite relation and endpoint attributes are simultaneously hidden, while global release counts can couple their completions. When attribute supports are independent, our problem collapses to ordinary weighted matching; that case is a baseline, not a novelty claim.

Classical cell-suppression work solves disclosure-control problems under linear table constraints by integer programming and branch-and-cut [12]. Our inverse object is different: released threshold states constrain latent micro-labels that also determine a hidden matching. Integer programming remains a general fallback, not an algorithmic novelty.

Resource-constrained matching and structured inference. An additive score floor introduces the same second resource studied in standard budgeted matching, which is NP-hard and admits PTAS/EPTAS algorithms [2, 10]. We do not assume those schemes transfer to mandatory core cover, and a generic score constraint or approximation scheme is not our contribution. Likewise, bounded-width constraint processing and bucket elimination are standard [9, 13]. Global side constraints must be compiled before width is assessed: rainbow matching remains APX-complete even when the underlying edge-coloured graph is a path [16]. Our algorithmic increment is an executable temporal state for this release-coupled world model, with active factor lifetimes, threshold caps, score-resource tracking, and endpoint witnesses.

Weak supervision and calibrated restrictions. Multiple-instance learning recovers instance-level structure from bag labels only through modeling assumptions [18]. In our application, node match outcomes do not nonparametrically identify partner-edge rankings, so we treat any edge model as a scorer. Split conformal calibration supplies finite-sample marginal coverage under exchangeability [17]. More directly, Cauchois et al. develop conformal prediction with weak supervision for structured outputs and instantiate bipartite perfect matchings, partial-matching feedback, and matching-oracle implementations [3]. Thus neither matching-level conformal

prediction nor a matching oracle inside calibration is new here. Our calibration markets reveal the complete matching; we use rank calibration as a safety layer for a release-coupled nonbipartite optimizer. Candidate and release-set eligibility remain separate assumptions.

Ride pooling and contextual composition. Shareability networks and dynamic assignment solve a forward allocation problem with observed requests [1, 22]. Chicago work predicts sharing and matching outcomes [19, 26], relates pooled demand to neighborhood disadvantage [25], and constructs proxy groups by spatiotemporal clustering [24]. Our flagship application asks an inverse measurement question: among a precisely defined set of realized two-transaction matches, which neighborhood-context compositions remain compatible with the suppressed partner key and the verified release operator? It does not infer rider socioeconomic status, preferences, durable ties, or an echo chamber.

3 PROBLEM AND QUERY SEMANTICS

Released records and roles. Let V index released records and let $G = (V, E)$ be a declared candidate supergraph. Core records $C \subseteq V$ belong to the target population; boundary-buffer records $B \subseteq V$ may be partners of core records; and context-only records $U = V \setminus (C \cup B)$ enter privacy counts but cannot be selected as partners. A matching $M \subseteq E$ must satisfy

$$\deg_M(i) = 1 \quad (i \in C), \quad \deg_M(i) \leq 1 \quad (i \in B), \quad (1)$$

and contains no buffer–buffer edge. This boundary rule lets a core record use a counterpart immediately outside the target window without forcing every buffer record into the target.

Latent attributes and release constraints. Each record i has a finite support $\mathcal{A}_i$ for a latent categorical attribute a_i . Supports may be singletons for released values or larger sets for coarsened values. In the implemented reference solver, $h(i)$ assigns each record to one release cell for this modeled attribute. The verified observation operator supplies rational count bounds

$$L_{cv} \leq \sum_{i:h(i)=c} \mathbb{I}\{a_i = v\} \leq U_{cv} \quad (c, v) \in \mathcal{K}. \quad (2)$$

More specific threshold or paired-suppression logic can be compiled into the same feasible-world representation from explicit LOW/HIGH release formulas. The compiler requires an operator identifier, an audit reference, and exact row-to-factor bindings; these are provenance inputs, not proof that the external mechanism is correct. We do not infer a converse suppression rule from missingness alone.

Feasible worlds and queries. A feasible world $F = (M, a)$ satisfies Eqs. (1)–(2), $a_i \in \mathcal{A}_i$, and any declared attribute-conditioned edge rules. Let $\mathcal{F}(O, G)$ denote the finite set of such worlds. For a symmetric pair function g , define the core-transaction query

$$Q_C(F) = \frac{1}{|C|} \sum_{\{i,j\} \in M} |\{i, j\} \cap C| g(a_i, a_j). \quad (3)$$

Core–core edges contribute twice and core–buffer edges once, so the denominator remains $|C|$. Same-bin composition uses $g(a_i, a_j) = \mathbb{I}\{b(a_i) = b(a_j)\}$; signed linear or regression queries replace g by the corresponding endpoint contribution.

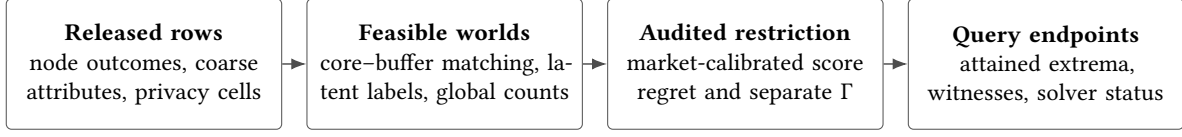


Figure 1: The method never fills in one preferred relation. The observation operator defines a joint world set; learning may restrict that set only through a separately calibrated rule; optimization then propagates all surviving ambiguity to the downstream query.

Table 1: Claim boundary induced by the four input layers.

Layer	Supplied object	Failure if misspecified
Relation support	Supergraph and core/buffer roles	True partner absent
Attribute support	Per-record finite domains	True endpoint value absent
Observation operator	Count and release factors	Inconsistent world deletion
Learned restriction	Score radius and Γ budget	Model-induced deletion

We report

$$\underline{Q} = \min_{F \in \mathcal{F}(O, G)} Q_C(F), \quad \bar{Q} = \max_{F \in \mathcal{F}(O, G)} Q_C(F). \quad (4)$$

PROPOSITION 1 (ENDPOINT SEMANTICS). *If $\mathcal{F}(O, G)$ is nonempty, both extrema in Eq. (4) are attained. Every feasible world’s query value lies between them. If the true world $F^\dagger$ belongs to $\mathcal{F}(O, G)$, its query value is covered. The proposition does not imply that every scalar between the endpoints is attainable.*

PROOF. The domain is a finite set of matchings and finite attribute assignments, so a real-valued query attains its minimum and maximum. The bracketing and coverage claims follow from membership. Two feasible worlds can attain only two distinct values, so finiteness alone provides no interpolation claim. $\square$

3.1 Observation Operators as Auditable Constraints

The world set must encode what the release mechanism actually implies, not a convenient completion rule. We separate four inputs. *Record supports* translate released or coarsened values into finite domains. *Count factors* constrain assignments across every row contributing to a privacy cell, including context-only rows. *Release factors* state the verified direction of the mechanism: for example, a visible value may imply that a count cleared a threshold, while a hidden value need not imply the converse. Finally, *compatibility factors* specify which attribute pairs allow an edge. An omitted rule leaves worlds in the set; it does not silently choose a default completion.

This separation yields three different statements. The score-free endpoints are conditional on the first three layers. A learned endpoint adds the fourth. A sampling statement additionally requires a design over complete markets. None of these is a confidence interval merely because two numbers are reported. The artifact

therefore stores the two endpoint witnesses and a solver-status record alongside every range.

4 COUPLED ENDPOINT COMPUTATION

The method must distinguish a useful baseline from the coupled problem.

LEMMA 1 (INDEPENDENT-SUPPORT COLLAPSE). *Suppose (i) E does not depend on latent attributes, (ii) the attribute domain is Cartesian, $\prod_i \mathcal{A}_i$, with no cross-record count constraint, and (iii) the query is pair additive as in Eq. (3). Define*

$$\ell_{ij} = \min_{a_i \in \mathcal{A}_i, a_j \in \mathcal{A}_j} | \{i, j\} \cap C | g(a_i, a_j),$$

$$u_{ij} = \max_{a_i \in \mathcal{A}_i, a_j \in \mathcal{A}_j} | \{i, j\} \cap C | g(a_i, a_j).$$

Then the lower endpoint is a minimum-weight core-cover matching with weights $\ell_{ij}/|C|$, and the upper endpoint is the corresponding maximum-weight matching with weights $u_{ij}/|C|$.

PROOF. For any fixed matching, selected edges share no endpoint. Under a Cartesian attribute domain, each selected edge’s minimizing or maximizing endpoint assignment can therefore be chosen independently and combined into one global assignment. Labels of unselected nodes can be completed arbitrarily from their nonempty supports. Optimizing the resulting edge sums over matchings gives the stated equalities. $\square$

Thus exact cover, independent missing-bin envelopes, and signed edge objectives are weighted-matching baselines. The nonseparable problem begins when release counts or attribute-conditioned feasibility prevent the edgewise choices in Lemma 1 from coexisting.

Why the joint world changes an endpoint. Consider four core records with fixed edges ab and cd . Their supports are $\{A, B\}$, $\{A, C\}$, $\{A, D\}$, and $\{A, E\}$, respectively. Let A form bin 0, B, D bin 1, and C, E bin 2. With independent supports, the same-bin query ranges from 0 to 1: choose (B, C, D, E) for the lower witness and all A for the upper witness. Add the single global release constraint $\#\{i : a_i = A\} \leq 2$. At most one edge can now have equal labels, so the attained range contracts to $[0, 1/2]$. Optimizing a matching range and an attribute range separately would retain the impossible upper value 1.

Attributes can also change the relation rather than only its query value. Let one core c have support $\{A, B\}$ and two singleton buffers $x = A$ and $y = B$. If edge cx permits only (A, A) and cy only (B, B) , a count constraint fixing $c = A$ selects x , while one fixing $c = B$ selects y . The implemented exact oracle reproduces both examples. They isolate the interaction absent from the independent-support

baseline without invoking a privacy mechanism that has not been verified.

A coupled complexity boundary. Fix an integer threshold $k \geq 3$. Consider a paired row-release operator that publishes both endpoint attributes of record i exactly when both its pickup and drop-off cells contain at least k records; otherwise it publishes neither. This is an abstract operator, not an assertion about the current Chicago release.

THEOREM 1 (PAIRED-SUPPRESSION FEASIBILITY). *For every fixed $k \geq 3$, deciding whether a feasible attribute assignment and perfect matching exist under the paired operator is NP-complete, even when the outer relation is complete bipartite, compatibility depends only on the left record's pickup attribute, and only left pickup attributes have non-singleton supports.*

PROOF SKETCH. Membership in NP follows by checking assignments, cell counts, release states, compatibility, and matching degrees. For hardness, reduce perfect three-dimensional matching on equally sized sets X, Y, Z . Each $x \in X$ chooses a latent pickup label $p_x \in Z$. For every z , add $k - 1$ visible context anchors to its pickup cell; all anchors share a separate already-large drop-off cell. Visibility forces $(k - 1) + |\{x : p_x = z\}| \geq k$. Since $|X| = |Z|$, every z is chosen exactly once. Give each x a private low-count drop-off cell so its two attributes remain paired-suppressed. Match X perfectly to Y and allow edge (x, y) under label z exactly when (x, y, z) is an input triple. Feasible worlds are therefore in bijection with perfect 3D matchings. The full construction, including the $|Z| \geq 2$ padding and size bound, is in Appendix B. $\square$

The reduction relies on paired row suppression and arbitrary tabular attribute-conditioned compatibility. The construction does not apply to endpoint-wise independent suppression and does not establish hardness for a metric-only edge rule. These boundaries must be checked, not blurred, in any application.

Compiling release formulas. For a scalar release factor f , let $N_f(a) = \sum_i \chi_{if}(a_i)$ with $\chi_{if} \in \{0, 1\}$. Its temporal automaton stores a capped count q_f and a requirement $\rho_f \in \{\emptyset, \text{LOW}, \text{HIGH}\}$.

For a pure threshold k , $q_f = \min\{k, N_f^{\text{processed}}\}$; closing the factor accepts LOW iff $q_f < k$ and HIGH iff $q_f = k$. Opposing nonempty requirements conflict. General lower/upper bounds use the smallest cap that retains every decision boundary, including a distinct overflow state for a binding upper bound.

For row i , substantive label a , and terminal factor predicates T_f , the implemented interface accepts an explicit DNF

$$\Phi_i(a) = \bigvee_{j=1}^{c_i(a)} \bigwedge_{(f,r) \in C_{iaj}} [T_f(N_f) = r], \quad r \in \{\text{LOW}, \text{HIGH}\}. \quad (5)$$

It replaces a by one compiled copy (a, j) per clause, copies all edge score, query, and compatibility maps, and emits the clause literals as factor requirements. The sole empty clause is canonical TRUE. Under the abstract paired operator, visible is one HIGH-HIGH clause and hidden is the two-clause LOW-or-LOW formula.

LEMMA 2 (EXPLICIT-DNF RELEASE COMPILER). *The projection of feasible compiled worlds is exactly the release-augmented world set*

satisfying Eq. (5). A projected world has one preimage for each combination of its satisfied clauses. Compilation preserves the node/edge topology, the record-bag width for the same forget order, score-floor feasibility, and every label- and edge-additive query; it changes label support from $|\mathcal{A}_i|$ to $\sum_{a \in \mathcal{A}_i} c_i(a)$ and may extend factor lifetimes and thereby increase active-factor width. Counts capped at their last decision boundary remain exact.

PROOF. For completeness, choose one satisfied clause for every row and attach its index to the substantive label. For soundness, read the clause embedded in each compiled label; accepting factor automata satisfy every emitted literal. The lifted maps copy feasibility, score, and query values across clause witnesses, so projection preserves endpoints. Compilation changes no record, edge, or action identifier. Counts on the same side of their final decision boundary have identical future transitions, so saturation is lossless. $\square$

The compiler is polynomial in the *explicit* DNF and lifted output, not a succinct arbitrary Boolean formula; exponential expansion and LOW/HIGH middle regions are exposed interface limits.

Temporal frontier. A supplied forget order is compiled into node-introduction, edge-processing, and node-forget actions. Every edge is processed while both endpoints are live; a record is forgotten only after all incident edges; and factor f is implicitly opened before its first possible contributor or requirer and closed immediately after its last. Let w be the maximum live-record count, d the largest compiled label support (including release-witness copies), B_t the live record bag, and $\mathcal{A}_t$ the active factor set at action t . Factor f retains counts $0, \dots, K_f$.

For allowed label-pair scores $s_e(u, v)$, set

$$a = \min \left\{ 0, \min_{e, (u,v) \in \mathcal{P}_e} s_e(u, v) \right\}.$$

Every world has total selected core incidence $N = |C|$, so replacing s_e by $s_e - a \geq 0$ adds the same $-aN$ to every score. Clear rational denominators and cap the accumulated integer score at the transformed floor b .

THEOREM 2 (EXACT SCORE-AWARE TEMPORAL-FRONTIER ENDPOINTS). *For a valid schedule, omission budget Γ , rational additive query, and score floor, the temporal-frontier dynamic program returns infeasible exactly when the declared world set is empty; otherwise it returns both attained query endpoints and witness worlds, provided the declared frontier-record limit is not exceeded. Exceeding that operational limit raises an explicit frontier-limit exception and licenses no feasibility conclusion. Before dominance pruning, the number of live keys at action t are at most*

$$\left(\prod_{i \in B_t} m_i d'_i \right) (\Gamma + 1)(b + 1) \prod_{f \in \mathcal{A}_t} 3(K_f + 1), \quad (6)$$

where $d'_i = \sum_{a \in \mathcal{A}_i} c_i(a)$, $m_i = 2$ for matchable records and $m_i = 1$ for context-only records. If $d'_i \leq d$, $|B_t| \leq w$, all active pure thresholds have cap k , and at most r are active, this is at most $(2d)^w [3(k + 1)]^r (\Gamma + 1)(b + 1)$. Without a score floor the factor $(b + 1)$ disappears. The work per transition is polynomial in the explicit state and input size.

PROOF SKETCH. At action t , a key stores live labels, live matching bits, the (q_f, ρ_f) automaton for each $f \in \mathcal{A}_t$, used omitted edges, and the score capped at b . For each key the program keeps separate exact minimum and maximum accumulated query values. These fields contain every processed quantity on which a future transition can depend. The running-intersection conditions ensure that forgotten labels and finalized factors can never be consulted again. Induction over actions therefore shows that the retained values are exactly the extrema among partial worlds inducing the key. Terminal paths have every core matched, every factor finalized, and score state b ; they project onto precisely the feasible full worlds. Stored choices recover the two witnesses. Counting each live label, matching bit, factor automaton, and global resource gives Eq. (6). $\square$

For a forget order $\pi = (v_1, \dots, v_n)$, define the forward boundary $\partial_\pi^+(i) = \{v_j : j > i, \exists h \leq i : \{v_h, v_j\} \in E\}$. The compiler's largest record bag is exactly $1 + \max_i |\partial_\pi^+(i)|$, the reverse-order vertex-separation number plus one; minimizing over orders yields pathwidth [15]. Hence, if records are ordered by nondecreasing time and every candidate edge spans at most Δ , then $w \leq \max_t \{i : t \leq \tau_i \leq t + \Delta\}$. Active-factor width is likewise the maximum overlap of the compiler-audited factor lifecycle intervals. These are data-measurable certificates for a supplied order, not a claim that the order minimizes pathwidth.

An exact algorithm polynomial in binary input length at fixed structural width would imply $P = NP$; our dynamic program pays a pseudo-polynomial score coordinate.

THEOREM 3 (WEAK HARDNESS AT PATHWIDTH TWO). *For non-negative integer score and query contributions encoded in binary, deciding whether a perfect matching has score at least B and additive query at most B is weakly NP-complete even with singleton labels, no release factors, $\Gamma = 0$, and a candidate graph that is a disjoint union of four-cycles.*

PROOF SKETCH. Reduce SUBSET SUM. For item a_i , a four-cycle has two perfect matchings, “take” and “skip”. Give one take edge score a_i per core incidence and query $2a_i$; all other contributions are zero. With both thresholds $2B$, a global matching is feasible exactly when the taken items sum to B . Each component, and their concatenation, has pathwidth two. Membership in NP is immediate. This is the model's pseudo-polynomial boundary, consistent with known budgeted-matching hardness and approximation results [2, 10]. $\square$

Certified score relaxation. When b is too large, choose rational $\eta > 0$, set $r_e = \lfloor (s_e - a)/\eta \rfloor$, and impose

$$\hat{b} = \max\{0, \lceil (B - Na)/\eta \rceil - N\}.$$

PROPOSITION 2 (OUTWARD SCORE CERTIFICATE). *If $\mathcal{F}_B$ is the exact score-floor set and $\hat{\mathcal{F}}_\eta$ the rounded set, then*

$$\mathcal{F}_B \subseteq \hat{\mathcal{F}}_\eta \subseteq \mathcal{F}_{B-\eta N}. \quad (7)$$

Consequently, optimizing over $\hat{\mathcal{F}}_\eta$ gives outward query endpoints. Each returned witness misses the original score floor by at most ηN ; if an endpoint witness satisfies the original floor, that endpoint is exact.

PROOF. Write $c_e = |e \cap C|$ for selected-edge core incidence and suppress the selected label-pair argument of s_e and r_e . Fixed core incidence gives $\eta \sum c_e r_e \leq \sum c_e (s_e - a) < \eta (\sum c_e r_e + N)$, which proves both inclusions after integer rounding. Set inclusion gives the endpoint order. If a relaxed extremal witness is also exact-feasible, the reverse endpoint inequality follows from that witness. $\square$

Equation (7) is a bicriteria score certificate, not a query FPTAS: a tiny score slack may cause an arbitrary query jump.

Reference formulation. The implemented coupled reference program uses binary edge variables z_e , assignment variables x_{iv} , and joint edge-label variables y_{euv} only for ordered pairs (u, v) in an edge's declared compatibility set $\mathcal{P}_e$. It enforces Eq. (1), $\sum_{v \in \mathcal{A}_i} x_{iv} = 1$, the count bounds in Eq. (2), and

$$\begin{aligned} \sum_{(u,v) \in \mathcal{P}_e} y_{euv} &= z_e, \\ y_{euv} &\leq x_{iu}, \quad y_{euv} \leq x_{jv}, \\ y_{euv} &\geq z_e + x_{iu} + x_{jv} - 2. \end{aligned}$$

The query is linear in y . Context-only records have assignment variables and enter counts but have no incident selectable edge. An exhaustive enumerator is the exact oracle on small instances; the sparse mixed-integer program is a numerical reference backend. It also enforces optional score floors and supergraph-edge budgets. Exact optima, numerical optima, exact or structural infeasibility, numerical infeasibility, and unresolved runs are reported as distinct states; a timeout is unresolved rather than a separate certificate.

The separate temporal implementation accepts tuple-valued compiled labels, multiple label-dependent count contributions and release requirements, and label-pair-specific scores and queries. It constructs sparse active-factor coordinates from a supplied forget order, uses exact rational arithmetic, and independently replays each endpoint witness. The high-level compiler implements Lemma 2, records support expansion, validates node/edge/factor lifecycles, and restores the selected DNF clause from a projected witness. Its mandatory audit reference does not verify an external operator, and no Chicago-specific null-cause adapter is claimed. The Python reference stores complete partial witnesses rather than memory-optimal predecessor pointers; Eq. (6) is its key-count bound, not a production runtime claim. General MILP remains the fallback when the measured temporal frontier is too wide.

5 LEARNING-AUGMENTED RESTRICTIONS

5.1 What Node Outcomes Do Not Identify

Let $\partial M \in \{0, 1\}^V$ be the incidence vector of a matching. A released match indicator can reveal ∂M without revealing M .

PROPOSITION 3 (PARTNER NONIDENTIFICATION). *If a candidate graph contains distinct matchings M_1 and M_2 with $\partial M_1 = \partial M_2$, then node match outcomes and node covariates alone do not identify which of those relations occurred. In particular, an all-positive incidence vector does not distinguish any two perfect matchings.*

PROOF. Take two latent worlds with the same released nodes, covariates, and incidence vector, but relation M_1 in one and M_2 in the other. Their observations coincide while their partner edges

Table 2: Algorithmic boundary of the declared world model.

Regime	Endpoint computation	Status
Fixed attributes	Weighted core-cover matching	Polynomial baseline
Independent supports	Two edge-envelope matchings	Lemma 1
Fixed labels plus score floor	Mandatory core cover with a budget	Related standard problem: NP-hard; PTAS/EPTAS known
Paired threshold coupling	Joint labels and matching	Theorem 1
Temporal schedule ($w, \mathcal{A}_t$)	Exact frontier DP	Theorem 2
General compiled graph	MILP/decomposition	Certified gap required

Table 3: Machine-readable solver states and their licensed interpretation.

State	Source	Licensed statement
EXACT_OPTIMAL	Exhaustive oracle or frontier DP	Attained endpoint certified
EXACT_INFEASIBLE	Frontier DP	Declared world set empty
OUTER_OPTIMAL	Rounded frontier DP	Outward endpoint; report score slack
NUMERICALLY_OPTIMAL	HiGHS incumbent	Numerical endpoint; retain gap
PROVEN_INFEASIBLE	Structural check or enumeration	Declared world set empty
NUMERICALLY_INFEASIBLE	HiGHS status	Numerical evidence only
UNRESOLVED	State/time limit or solver failure	No feasibility conclusion

differ. No function of the released data can distinguish them without another model or source of relation truth. $\square$

This result does not prohibit structured weak supervision; it fixes its claim boundary. A model may score complete matchings and enforce mutual exclusion, but a node-level objective alone cannot validate partner ranks in the regime of Proposition 3. In our controlled benchmark, edge scorers are therefore trained on separate synthetic source markets whose relations are observed. Chicago node outcomes are not used as hidden edge labels.

5.2 Calibrating a Matching Set

Let s_e be a matching-only scorer fixed and deterministically rationalized before calibration. With optional buffer nodes, use core incidence,

$$S_C(M) = \sum_{e \in M} |e \cap C| s_e. \quad (8)$$

Because $\sum_{e \in M} |e \cap C| = |C|$, replacing s_e by $as_e + b$ for $a > 0$ changes every feasible score by the same additive constant after scaling. For

market d , let $h_d(M)$ count selected edges in $E_d^{\text{super}} \setminus E_d^{\text{base}}$ and define

$$\mathcal{F}_{d,\Gamma}^0 = \{F \in \mathcal{F}(O_d, G_d^{\text{super}}) : h_d(M(F)) \leq \Gamma\}.$$

Before calibration, fix a reference budget $\bar{\Gamma}_d$ by the same rule in every market and compute once

$$S_{d,\star}^- = \min_{F \in \mathcal{F}_{d,\bar{\Gamma}_d}^0} S_C(M(F)), \quad S_{d,\star}^+ = \max_{F \in \mathcal{F}_{d,\bar{\Gamma}_d}^0} S_C(M(F)).$$

These extrema are frozen along the entire Γ path. Define

$$R_{d,\star}(F) = \frac{S_{d,\star}^+ - S_C(M(F))}{S_{d,\star}^+ - S_{d,\star}^-}, \quad (9)$$

with value zero for a constant reference score. Recomputing the normalizer at each Γ can destroy nesting: a newly admitted high-score world can raise the floor enough to delete an old world.

For any $\tau \in [0, 1]$ and $\Gamma \leq \bar{\Gamma}_d$, define

$$\mathcal{F}_{d,\tau,\Gamma} = \{F \in \mathcal{F}_{d,\Gamma}^0 : S_C(M(F)) \geq S_{d,\star}^+ - \tau(S_{d,\star}^+ - S_{d,\star}^-)\}. \quad (10)$$

Use m calibration markets, disjoint from scorer training, whose true matchings are observed. For score miscoverage α_S , let $k_S = \lceil (m+1)(1-\alpha_S) \rceil$ and let $\widehat{\tau}_{\alpha_S}$ be the k_S th smallest calibration regret, using the support bound one if $k_S = m+1$. Retain the set in Eq. (10) at $\tau = \widehat{\tau}_{\alpha_S}$. Exact additive scores parse each component through its declared decimal spelling before rational termwise summation; the same score map is used for calibration and the endpoint solver. A label-dependent scorer would instead require calibration truth for the full world, not only its matching.

THEOREM 4 (MARKET-LEVEL SCORE RETENTION AND QUERY COVERAGE). *Suppose the scorer, observation operator, graph rule, reference budgets, and normalization are fixed before calibration. Suppose the augmented markets, including their true worlds, are exchangeable, every reference set is nonempty, and $F_d^\dagger \in \mathcal{F}_{d,\bar{\Gamma}_d}^0$ almost surely. Then*

$$\Pr\{R_{m+1,\star}(F_{m+1}^\dagger) \leq \widehat{\tau}_{\alpha_S}\} \geq 1 - \alpha_S.$$

For a reported $\Gamma \leq \bar{\Gamma}_{m+1}$, if $\Pr\{F_{m+1}^\dagger \notin \mathcal{F}_{m+1,\Gamma}^0\} \leq \alpha_G$, then

$$\Pr\{F_{m+1}^\dagger \in \mathcal{F}_{m+1,\widehat{\tau}_{\alpha_S},\Gamma}\} \geq 1 - \alpha_S - \alpha_G.$$

On this event, every query range optimized over that same set covers its true value.

PROOF. Exchangeability gives the usual split-conformal rank bound. A union bound with the final- Γ eligibility failure gives the second display; independence is unnecessary. Conditional on true-world retention, Proposition 1 covers all queries sharing that world set, so no union bound over queries is needed. $\square$

A supplied supergraph or synthetic calibration alone establishes neither rate for Chicago. With the frozen reference range, the two sensitivities do form a nested audit rather than one opaque knob.

PROPOSITION 4 (NESTED SENSITIVITY PATHS). *If $\tau \leq \tau'$ and $\Gamma \leq \Gamma' \leq \bar{\Gamma}_d$, then $\mathcal{F}_{d,\tau,\Gamma} \subseteq \mathcal{F}_{d,\tau',\Gamma'}$. Consequently, a query's lower endpoint cannot increase and its upper endpoint cannot decrease when either budget is relaxed.*

PROOF. Both structural budgets and one fixed score halfspace are nested. Optimization over the resulting superset gives the endpoint monotonicity. $\square$

Table 4: Locked temporal-frontier checks; timings are not headline evidence.

Family	Controlled change	Certified outcome
Local factors	Grouped vs. interleaved	active $r : 1 \rightarrow 3$; same endpoints
Degree-one graph	Adjacent vs. nested order	schedule width $1 \rightarrow 16$
Binary C_4 items	pathwidth $2, b = 2048$	peak frontier 4098
Γ path	$0 \rightarrow 8$ omitted edges	upper endpoint $0 \rightarrow 8$
Score relaxation	$\eta = 1/5, N = 4$	$[0, 0] \subset [-5, 0]$; deficit $2/5 \leq 4/5$

Reporting the (τ, Γ) path exposes whether apparent precision comes from score or candidate exclusion; it does not turn either into graph-support evidence.

6 CONTROLLED EVALUATION

6.1 Structural and Solver Audit

The locked temporal benchmark contains 34 hashed cases. An implementation-independent rational enumerator agrees in 32/32 applicable cases, analytic answers agree in 24/24, and resolved HiGHS comparators agree numerically in 16/16. Every endpoint witness is replayed from raw inputs. Table 4 isolates the structural parameters rather than presenting wall-clock speedups. A separate legacy audit of the general MILP backend still agrees with exhaustive statuses on 250/250 random instances and endpoints on all 185 feasible instances; that agreement does not make floating-point MILP exact.

The artifact has 96 passing unit tests and eight executable legacy counterexamples. The temporal suite includes 80 seeded brute-force instances; the relaxation suite adds random inclusion and witness checks. These results support software behavior for declared finite inputs, not candidate coverage, a production schedule, or the correctness of an external release operator.

A bounded 23-case capacity profile varies all six state-bound axes. Twenty-two cases resolve under three seconds; the degree-five 14-record case remains an explicit timeout. Peak frontiers grow $64 \rightarrow 512$ with zero-to-three overlapping factors and $8 \rightarrow 1024$ with support one-to-four. These machine-specific diagnostics certify only the supplied heuristic orders, not production scale.

6.2 External Relation-Truth Boundary

UCI Krebsregister block 1 has 574,913 candidate pairs and 2,093 adjudicated links [23], but its truth is not a matching: 152 records have degree above one. We preserve that failure. A disclosed truth-conditioned reduction of 1,796 dyads gives a numerical postal-agreement frontier $[0.918151, 0.954900]$, with truth at the upper endpoint. Complementary FEBRL4 tests [20] cover truth on a six-pair exhaustive and a 20-pair numerical complete-matching market. These tests establish topology boundaries and method fit, not UCI blocking recall, exchangeability, calibrated learning, or Chicago transfer.

6.3 Matching-Set Calibration under Deliberate Query Leakage

Design. We use a deterministic, disjoint source/calibration/test split. Forty source markets directly supervise logistic edge scorers with observed true edges; no node-label learner is evaluated here. Forty-nine target-style markets calibrate Eq. (9), and 120 held-out markets are used for final evaluation. Each market has six hidden pairs, fully observed binary endpoint attributes, and a complete candidate graph. There is no privacy-count coupling in this experiment, so it isolates the matching-set calibration layer.

Target-free features measure route, time, and duration similarity. A query-leaking diagnostic additionally sees the same-SES edge contribution that defines the downstream query. The source homophily generating probability is 95%, while calibration and test use 55%, deliberately shifting that circular feature. The nominal matching-set coverage is 90%. Radius 0.05 is an illustrative stress point, not a selected or calibrated baseline. The committed benchmark enumerates all 10,395 perfect matchings per market using termwise decimal-rational scores and exact floor membership. A second solver-free program shares the locked generator, scorers, and rationalization utilities but separately reaggregates the 44-point frontier: discrete counts agree exactly and serialized headlines within 3.33×10^{-16} . It scores each edge once; the all-core incidence convention in Eq. (8) doubles every matching score and leaves normalized-regret sets unchanged.

Result. The query-leaking radius-0.05 stress point looks strongest if evaluated only by width: it removes 95.8% of the score-free ambiguity. It retains 18/120 true matchings and covers the downstream query in 29/120 markets. Calibration detects the shifted shortcut. Its radius expands to 0.504, retaining 117/120 true matchings and covering the scalar query in 120/120 markets, while width reduction falls to 7.6%. The target-free score needs a smaller radius but also gains little calibrated precision. This single seed and market size do not establish robustness. The result is narrower: calibration makes the coverage price of sharpening measurable rather than rewarding circular precision.

Software audit. The benchmark validates matching-level calibration only. It does not validate node weak supervision, latent-attribute inference, privacy-count coupling, candidate support, or transfer to Chicago. Multi-seed, size, sparsity, coarsening, and candidate-omission sweeps remain required evidence.

7 CHICAGO FLAGSHIP APPLICATION: EVIDENCE GATE

Chicago’s public Transportation Network Provider table reports whether each customer transaction shared a vehicle with a separately booked customer and reports the pooled run size [7, 8]. The provider-side Shared Trip ID groups the empty-to-empty vehicle run but is absent from the public schema. Conditional on run closure and reporting consistency, records with match true and $K = 2$ therefore induce a hidden nonbipartite transaction matching. This is neither a passenger count nor a claim that $K > 2$ runs form co-presence cliques.

The current base dataset metadata points to the City’s privacy methodology as the approach used in this release. That note lists the

Table 5: Fixed held-out synthetic markets: coverage cannot be read from width alone.

Scorer	Calibrated restriction			Arbitrary radius 0.05			Point MAE
	Match cov.	Query cov.	Width red.	Match cov.	Query cov.	Width red.	
Target-free	95.8%	100%	3.8%	30.0%	83.3%	70.6%	0.122
Query-leaking diagnostic	97.5%	100%	7.6%	15.0%	24.2%	95.8%	0.308

The score-free range covers the query in every test market and has mean width 0.661. Calibrated radii are 0.341 (target-free) and 0.504 (query-leaking). “Match cov.” tests whether the entire hidden matching remains eligible; “Query cov.” is weaker because an excluded matching’s scalar value can still fall between the endpoints. The observed 100% values mean 120/120, not a strengthened theoretical coverage level.

current dataset and documents an at-most-two-unique-trip threshold for 15-minute tract cells with paired coarsening of both endpoints [6]. A companion clarification describes separate pickup/start and drop-off/end cells [5]. We therefore compile the safe one-way implication: a published fine tract rules out a low-count applicable internal cell at either endpoint. We do *not* infer LOW from a blank tract; the City also lists missing source geography and outside-city locations. Every trip contributing to a relevant cell, not only the matched cohort, must enter the count factors. Documentary support is not an audit of the City’s code, provider partitioning, DST handling, or late-row recomputation.

The production analysis is still not licensed. The committed data are a deterministic two-character trip-ID-prefix sample, which omits a retained record’s counterpart in approximately 255 out of 256 cases under independent prefix retention. Even an authorized-only complete day does not prove run closure: a partner can cross midnight or released flags can be inconsistent. The City may publish after generally 99% or more expected data have arrived and later revise past periods [4]; hence “complete day” means complete only within a pinned snapshot and requires a later stabilization check.

The complete application requires all transactions contributing to each privacy cell, run-closed boundary buffers, a necessary-condition candidate supergraph, the correct tract vintage, and full solver status by component. Primary comparisons will include complete cases, community-area aggregation, independent attribute envelopes, one point imputation, spatiotemporal proxy grouping, score-free coupled endpoints, and calibrated restrictions only where independent partner truth supports them. No Chicago composition or January 2026 policy sign is reported in this draft.

8 LIMITATIONS AND ETHICAL CONSIDERATIONS

The endpoint guarantee is conditional on the declared world set. Candidate omissions, an incorrect release operator, nonexchangeable calibration markets, or unmodeled attribute supports can remove truth before optimization. The split-conformal result is marginal across complete markets, not conditional on every market and not transferable from simulation by assertion. Numerical MILP optima are not exact mathematical certificates; production reporting must retain primal/dual gaps and unresolved states.

The Chicago unit is a customer transaction, not a stable person. Census attributes describe neighborhood context, not rider income or race. Releasing one extremal latent assignment could facilitate reidentification, so the intended artifact reports only aggregate endpoints, support diagnostics, and solver certificates. It must not

publish imputed partner links, trip IDs, or latent fine-geography assignments. Institutional review and the City’s data terms must be resolved before any nonpublic data are used.

9 GENERATIVE AI USAGE

Generative AI assisted code drafting, adversarial counterexample search, and language editing. Every mathematical claim, experiment, citation, and code path remains subject to author verification. No generated text or synthetic output is treated as evidence about Chicago.

10 CONCLUSION

Privacy coarsening can hide a relation and its endpoint attributes at the same time. The appropriate output is then a query over feasible worlds, not a preferred reconstructed graph. Independent attribute supports reduce to weighted matching; active release factors create the coupled problem. The temporal-frontier algorithm gives exact endpoints when live structural width and the score resource are controlled, while the outward relaxation remains honest when the exact score frontier is too large. Matching-level conformal calibration is a known safety layer that can expose false precision, not the algorithmic contribution. The remaining work is decisive: verify and compile the City implementation against complete snapshots, construct and measure a run-closed full-day temporal order, scan the full external corpus, and run the Chicago application.

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A REPRODUCIBILITY AND CLAIM STATUS

The artifact includes the complete benchmark generator, split-conformal calibrator, exact temporal-frontier solver, small-instance matching oracle, sparse numerical reference solver, fixed seeds, result tables, and executable adversarial examples. Hidden pair truth is read only in the source-training, calibration, or held-out-test stage declared by the benchmark split. The earlier two-day weak-node-score pilot is retained as failure-analysis provenance but is not a headline result.

Current completed claims are limited to Propositions 1, 2, 3, and 4, Lemmas 1 and 2, Theorems 1, 2, 3, and 4, 96 passing declared-input unit tests, eight legacy counterexamples, the 34-case temporal benchmark, the seeded 250-instance MILP audit, and the controlled results in Table 5. The explicit-DNF LOW/HIGH release compiler, 23-case bounded capacity profile, UCI block-1 boundary audit, and FEBRL4 method-fit audit are also complete under their stated contracts. A Chicago null-cause adapter, run-closed full-day order, all-ten-block UCI scan, and complete-day Chicago estimates remain open gates.

B FULL PROOF OF THEOREM 1

Define the decision problem as follows. Records have pickup and drop-off time labels, finite attribute supports, and an observed symbol that is either a visible attribute pair or missing. A world chooses both attributes. The paired operator releases both exactly when both selected endpoint cells have count at least k . Target records form left and right sets and must be perfectly matched; context records cannot be matched. An explicitly represented table $\Gamma_{xy}(p_x)$ determines compatibility. A certificate is polynomially checked by computing all counts and matching degrees, so the problem is in NP.

Reduce perfect 3D matching with $|X| = |Y| = |Z| = q$. If needed, add one disjoint forced triple so $q \geq 2$ without changing the answer. Create target records r_x and r_y and the complete outer graph $X \times Y$. All r_x share one pickup time and have support Z . Each r_x has a private drop-off time and a singleton drop-off support; its observed pair is missing. Each r_y has private endpoint times, singleton supports, and a missing observed pair.

For each $z \in Z$, create $k - 1$ context anchors with the common pickup time and singleton pickup support $\{z\}$. All anchors share a drop-off time and a singleton label unused by targets, and are visibly released. The anchor drop-off count is $q(k - 1) \geq k$. If $n_z = |\{x : p_x = z\}|$, each z -anchor pickup count is $(k - 1) + n_z$; visibility is therefore equivalent to $n_z \geq 1$. Because $\sum_z n_z = q$, every $n_z = 1$. Missing target outputs are consistent: every x has a unique count-one drop-off cell, and every y has unique count-one endpoint cells.

Finally set $\Gamma_{xy}(z) = 1$ exactly for input triples (x, y, z) . A feasible world matches every x and y once and assigns every z once, so its selected triples form a perfect 3D matching. Conversely, a perfect 3D matching selects the corresponding p_x and (x, y) edges; all anchor counts equal or exceed k , all targets remain suppressed, and every selected edge is compatible. For fixed k the construction has $O(q)$ records, $O(q^2)$ support and outer-edge entries, and at most an $O(q^3)$ explicit compatibility table. The reduction is polynomial.

C PROOF DETAILS FOR THEOREM 2

Let X_t be a live key after action t and let $\mathcal{W}_t(X_t)$ be the partial compiled worlds that induce its live labels and matching bits, active capped counts and merged requirements, used omission budget, and capped score. They have already satisfied every forgotten-core degree constraint and every closed factor. For endpoint sense $\circ \in \{\min, \max\}$, the invariant is

$$V_t^\circ(X_t) = \bigcirc_{F \in \mathcal{W}_t(X_t)} H_t(F),$$

with the value attained by the stored partial witness.

The empty layer satisfies the invariant. A node introduction enumerates every compiled label, increments precisely its active factor automata, merges its requirements, and rejects exactly conflicts or factor states with no feasible completion. An edge action enumerates skip and select; select is legal exactly when both matching bits are zero and the live labels form an allowed pair, and it updates the two global resources. A forget action checks a core's bit and removes data whose final incident edge and factor have already been processed. Closing a factor applies its terminal predicate and removes its automaton. Thus every transition is a legal extension and every legal extension is enumerated.

By the schedule's running-intersection property, two partial worlds with the same key have identical future extensions and future additive contributions. Keeping only their attained minimum and maximum is therefore lossless. Induction proves the invariant at every layer. Terminal accepting paths have all core degrees one, all factors closed, omission count at most Γ , and capped score b ; conversely every feasible full world traces such a path. Backtracking yields the two endpoint witnesses.

For live record i , there are $m_i d'_i$ label/bit assignments; active factor f has at most $3(K_f + 1)$ automaton states, alongside $\Gamma + 1$ omission and $b + 1$ capped-score states. Their product gives Eq. (6); each explicit transition performs polynomial work. The reference implementation copies complete partial witnesses for auditability, whereas a predecessor-pointer implementation attains the usual rolling-state storage bound.

D FULL PROOF OF THEOREM 3

Membership in NP follows by checking a proposed perfect matching and its two integer sums. Reduce SUBSET SUM with positive integers $a_1, \dots, a_m$ and target B . For each item create a cycle $u_i - v_i - x_i - y_i - u_i$ with all four vertices core. Its only perfect matchings are

$$M_i^{\text{take}} = \{u_i v_i, x_i y_i\}, \quad M_i^{\text{skip}} = \{v_i x_i, y_i u_i\}.$$

Assign $s_{u_i v_i} = a_i$ and zero score to the other three edges. Since the edge is core-core, taking contributes $2a_i$ to the core-incidence score. Give the same take edge query $2a_i$, with every other query zero. A global perfect matching independently chooses take or skip in each cycle and hence has score and query both equal to $2 \sum_{i \in I} a_i$. Requiring score at least $2B$ and query at most $2B$ is therefore equivalent to $\sum_{i \in I} a_i = B$.

Each cycle has the width-two path decomposition

$$\{u_i, v_i, y_i\}, \quad \{v_i, x_i, y_i\}.$$

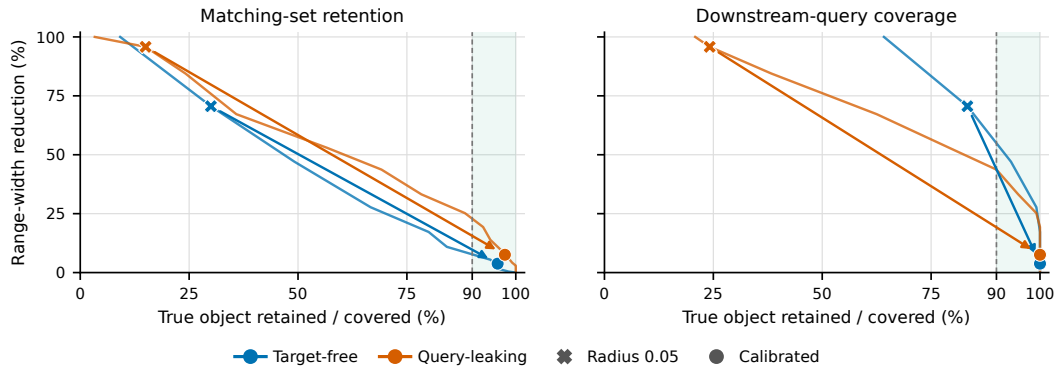


Figure 2: Exact radius frontiers show that the tight stress point has large width reduction but poor retention/coverage; calibration crosses the nominal 90% line while removing most apparent precision. Arrows connect radius 0.05 to the calibrated radius.

Decompositions of disjoint components can be concatenated without increasing width. The associated decision problem is pseudo-polynomially solvable by the score-frontier DP, so it is weakly NP-complete.